THE DERIVATIVE · TOPIC 1

Differentiating sin, cos, tan, e^x and $\ln x$

Thirty questions on the trigonometric, exponential and logarithmic derivatives, in three tiers of ten.

Pure Mathematics 1, 7.3–7.4 · Pure Mathematics 2, 4.1–4.4 · 30 questions · 60 marks

KEY WORDS

- radian
- chain rule
- product rule
- quotient rule
- natural logarithm
- exponential function
- logarithmic differentiation
- stationary point
- second derivative

REMEMBER

1. $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$.
2. $(e^x)' = e^x$ and $(\ln x)' = \dfrac{1}{x}$.
3. Chain rule: differentiate the outside, then multiply by the derivative of the inside.
4. $\dfrac{d}{dx}\ln f(x) = \dfrac{f'(x)}{f(x)}$.
5. Radians throughout. Show the rule you are using — method carries marks even when the answer slips.

Easy

EASY · 10 questions · One line from the table. 1 mark each.

1 Differentiate $y = 5\sin x$.

2 Differentiate $y = \cos 2x$.

3 Differentiate $y = e^{5x}$.

4 Differentiate $y = \ln 3x$.

5 Differentiate $y = \tan 4x$.

6 Differentiate $y = x + \sin x$.

7 Differentiate $y = 3e^{x} - 2 \ln x$.

8 Differentiate $y = \sin(x + 2)$.

9 Differentiate $y = e^{-x}$.

10 Differentiate $y = 4\cos x + \tan x$.

Medium

MEDIUM 10 questions

A chain, a product or a quotient. 2 marks each.

1 Differentiate $y = \cos^{3}x$.

2 Differentiate $y = x^{2}e^{x}$.

3 Differentiate $y = \ln(\sin x)$.

4 Differentiate $y = e^{x^2}$.

5 Differentiate $y = \sqrt{\sin x}$.

6 Differentiate $y = \sin x \cdot \cos x$ and simplify.

7 Differentiate $y = \dfrac{e^x}{x}$.

8 Differentiate $y = x \ln x - x$.

9 Differentiate $y = \tan^2 x$.

10 Differentiate $y = (\ln x)^3$.

Hard

HARD 10 questions

Several steps, or an idea. 3 marks each.

1 $y = e^x \cos x$. Find $\dfrac{d^2y}{dx^2}$.

2 Differentiate $y = \ln \dfrac{1 + x}{1 - x}$ and simplify.

3 Differentiate $y = x^{\ln x}$ for $x > 0$.

4 Find the tangent to $y = e^x$ that passes through the origin.

5 Find and classify the stationary point of $y = xe^{-x}$.

6 Show that $y = e^{2x}\sin 3x$ satisfies $y'' - 4y' + 13y = 0$.

7 Differentiate $y = \ln(x + \sqrt{x^2 + 1})$ and simplify.

8 Find the least value of $y = e^x + e^{-x}$.

9 Differentiate $y = \dfrac{1 + \tan x}{1 - \tan x}$ and simplify.

10 Prove that $(\sec x)' = \sec x \cdot \tan x$, starting from $\sec x = \dfrac{1}{\cos x}$.